Wavefront MAT: Design and Specification

Version 2.0 (draft for review) · 11 October 2026 · michael

Revision history

Version	Date	Status	Changes
2.0	2026-10-11	draft for review	holes: a region is any number of disjoint, nested loops (SC-01, IN-02, IN-05 withdrawn, IN-12 to IN-14, ALG-07); loop merges and closed plateaus (EV-09, EV-11, EV-14, RB-03, RB-07); the MAT of each component has one cycle per hole (OUT-08, OUT-11); API and JSON for several loops (API-02, OUT-06); relaxed verification scope (PF-03, VER-02 to VER-05); milestone M4 in three parts
1.5	2026-10-10	minor revision	OUT-05, API-02: Bézier export as a list of arcs, each turning by less than a right angle, and the flattening rule; OUT-06: the MAT JSON layout; ALG-06, API-02: `inward_offset` as implemented in M7; API-03: CLI options; section 13: the PF-02 targets move from M6 to a new milestone, M8
1.4	2026-10-09	minor revision	EV-11, EV-12, RB-01: the windowed broad phase as implemented in M6 (window choice, split candidates, event identity in the queue order); PF-01: heap compaction rule; RB-03: tangent ports compared within the cluster's spread, fast shocks die if they pass through D within ε_t; PF-02: times measured in M6; VER-07: benchmark families
1.3	2026-10-09	minor revision	EV-04: a circle front can also reach a regular vertex; RB-02, RB-03: cluster radius, port ordering and vertex kinds as implemented in M5; RB-04 and API: `resolve_all_events` option for the cross-check
1.2	2026-10-09	minor revision	API-03: error code `unsupported` for valid input that a later milestone handles
1.1	2026-10-09	minor revision	VER-01 and M1: kernel fixtures generated with SymPy instead of Maple
1.0	2026-10-09	approved baseline	content of 0.2 approved by michael as the implementation baseline; sources moved to `docs/` in the wfmat repository, which is now the master copy; tagged `spec-v1.0`
0.2	2026-10-09	preliminary draft	requirement identifiers added to every normative statement; requirements index (appendix A); tests cite the IDs they verify
0.1	2026-10-09	preliminary draft	first complete draft; scope decisions of 2026-10-08 applied (single loop, interior MAT, C++20, bulge JSON input)

Requirement identifiers. Every normative statement carries an identifier of the form **[AREA-nn]**, where AREA is one of SC (scope), IN (input), OUT (output), ALG (algorithm), EV (events), KN (kernel), RB (robustness), PF (performance), API, LIB (libraries and build) and VER (verification). Identifiers are permanent: a withdrawn requirement keeps its number,

marked withdrawn, and numbers are never reused. Every test names the identifiers it verifies as Catch2 tags, for example `[IN-04]`. Appendix A lists all identifiers.

1. Overview and scope

We compute the exact medial axis transform (MAT) of a planar region bounded by line segments and circular arcs by simulating the inward grassfire: the boundary offset front is propagated as a kinetic data structure, and every MAT edge is the trajectory of a front corner (a shock). Time equals offset distance, so the radius function falls out for free: a point traced at time t has radius $r = t$.

Goals

- **[SC-01]** Input: one or more closed loops of line segments and circular arcs, pairwise disjoint and nested to any depth, as C++ objects or a JSON file. The region is the material between each outer loop and the loops directly inside it (its holes); a loop inside a hole is the outer loop of another, separate component (an island).
- **[SC-02]** Output: the full interior MAT as a planar graph. Each edge is an exact conic arc (line, parabola, ellipse or hyperbola branch) with closed-form position $p(r)$, radius r, and the two boundary contact points as functions of r.
- **[SC-03]** Exact handling of the degeneracies that dominate CAD data: parallel lines, concentric arcs, tangent (G^1) joins, cocircular contacts and symmetric shapes.
- **[SC-04]** C++20, permissively licensed dependencies only, deterministic results, no global state.

Non-goals for v2

- The exterior MAT (outside the outermost loops or inside a hole), free-form curves (splines, NURBS), external file readers such as DXF, 3D, and pruning or simplification of the axis (offered later as post-processing on the output graph).
- Loops that touch or cross one another (a hole touching its outer loop, two holes sharing a point).
- Large-scale performance work: v2 is verified for correctness up to about 5,000 segments (PF-03).
- Exact (algebraic) arithmetic. We use double precision with tolerance-controlled topology and extended-precision refinement of event times (section 8).

Why wavefront. Unlike an incremental Voronoi construction, the front simulation yields MAT edges already ordered by radius, produces offset curves as a by-product, and handles arcs natively because offsets of lines and circles stay lines and circles. Its weak points are event detection for non-adjacent front pieces and simultaneous events; sections 6 to 8 address both.

2. Mathematical model

Every boundary piece becomes a site with a signed distance function that is linear (lines) or a shifted Euclidean norm (circles), so every bisector is a conic and every event is the root of at most a quadratic.

Region and MAT. Ω is a bounded open region with boundary $\partial\Omega$, and $d(x) = \operatorname{dist}(x, \partial\Omega)$. The MAT is the pair (M, r): M is the closure of the set of centres of maximal inscribed disks, and $r(x) = d(x)$ on M. M is a finite graph homotopy-equivalent to Ω: one connected piece per component of Ω, and in a component with h holes exactly h independent cycles, each enclosing one hole. Without holes it is a tree.

Sites. Each input segment is a site. Each reflex corner (a right turn when walking a boundary loop with Ω on the left; on a hole that is a corner pointing into the material) adds a point site at the corner. For x in a site's zone, its distance function is

$$d_L(x) = n \cdot x - c, \qquad \|n\| = 1,\ n \text{ pointing into } \Omega \qquad \text{(line)}$$

$$d_C(x) = \sigma\left(R - \|x - c\|\right), \qquad \sigma = +1 \text{ convex},\ \ \sigma = -1 \text{ concave} \qquad \text{(circle)}$$

A convex arc has its centre on the Ω side; a concave arc bulges into Ω; a point site is a concave circle with $R = 0$.

Front. $F(t) = \{\, x \in \Omega : d(x) = t \,\}$. Each front element is the live part of an offset site: a line $n \cdot x = c + t$, or a circle of radius $R - \sigma t$ about the same centre. A convex circle element dies no later than $t = R$.

[ALG-01] Corner classification. At each junction of consecutive segments a and b, let φ be the signed turning angle from the end tangent of a to the start tangent of b, with Ω on the left.

Turning angle φ	**Corner**	**Front vertex at $t = 0$**
$\varphi > \varepsilon_{\text{ang}}$	convex	shock vertex; a MAT edge starts at the corner with $r = 0$
$\lvert\varphi\rvert \le \varepsilon_{\text{ang}}$	tangent (G^1)	regular vertex; moves along the shared normal, traces nothing
$\varphi < -\varepsilon_{\text{ang}}$	reflex	point site inserted between a and b, flanked by two regular vertices
$\lvert\varphi\rvert \ge \pi - \varepsilon_{\text{ang}}$	cusp	excluded by the input contract; validation rejects it as a safeguard

Bisectors. A shock vertex between sites A and B traces the locus $d_A = d_B$, a conic:

Site pair	**Bisector**	**Remarks**
line, line	line	angle bisector; parallel facing lines give a plateau (constant r)
line, circle (any σ, incl. point)	parabola	focus at the circle centre
convex, concave circle	ellipse	foci at both centres, $\|x - c_1\| + \|x - c_2\| = R_1 + R_2$
convex, convex or concave, concave	hyperbola branch	$\|x - c_1\| - \|x - c_2\| = R_1 - R_2$; a line when $R_1 = R_2$
point, point	line	perpendicular bisector

Kinematics. Every regular point of the front moves along its normal at unit speed. A shock vertex whose front corner has interior angle α moves at speed $1/\sin(\alpha/2)$. Hence the front is 1-Lipschitz in time in the Hausdorff metric: two front pieces at distance δ at time t cannot meet before $t + \delta/2$. This bound drives non-local event scheduling (section 6).

MAT vertex kinds. Corner (degree 1, $r = 0$, at a convex boundary corner); curvature end (degree 1, $r = R$, at a convex arc centre); junction (degree ≥ 3, three or more contacts); transition (degree 2, a contact passes from a site to its tangent or reflex neighbour, so the conic type changes); extremum (degree 2, a local minimum of r where two fronts first touch, or a local maximum where two shocks annihilate).

3. Input specification

The input is a region given as one or more closed chains of lines and arcs in bulge form; the loops are validated, their nesting is derived, each is oriented from its nesting depth, and the whole region is normalised before any propagation starts.

[IN-01] Representation. A loop is a cyclic list of vertices (x, y, β) with bulge β. The segment from vertex i to $i+1$ is a line when $\beta = 0$, otherwise a circular arc with $\beta = \tan(\theta/4)$, θ the signed sweep (the DXF LWPOLYLINE convention). Bulge form guarantees closure and G^0 continuity by construction. A single full circle is a loop of two half-arcs ($\beta = \pm 1$ each). An explicit (centre, radius, start angle, sweep) form is also accepted and converted.

[IN-02] JSON schema (v2)

```
{
  "units": "mm",
  "loops": [
    [[0,0,0],[100,0,0],[100,40,1],[60,40,0],[0,40,0]],
    [[20,15,1],[30,15,1]]
  ]
}
```

`loops` lists every loop in any order and with any orientation; the second loop above is a circular hole of diameter 10. The v1 form, `"outer": [...]`, stays valid and means a region of that one loop; it may be followed by `"holes": [[...], ...]`, which are simply further loops. A document must use `loops` or `outer`, not both. Loops are numbered from 0 in document order (`outer` first, then `holes`), and diagnostics name a loop by that number and a segment by its index in the loop.

Validation rules (reject with a diagnostic naming the loop and segment)

1. **[IN-03]** Every segment has length $> \varepsilon_{\text{len}}$; every arc has $|\theta| \in (\varepsilon_{\text{ang}}, 2\pi)$.
2. **[IN-04]** Each loop is simple: no segment pair intersects except consecutive segments at their shared vertex. Checked with an R-tree broad phase and exact line/arc intersection tests.
3. **[IN-05]** *Withdrawn in 2.0* (v1: exactly one loop). Replaced by IN-12 and IN-13.
4. **[IN-06]** No cusp corners (section 2 table). The data contract excludes them; the check is a safeguard.
5. **[IN-12]** The loops are pairwise disjoint: no segment of one loop intersects or touches a segment of another, so the distance between any two loops exceeds ε_{len}. Checked in the same R-tree pass as IN-04, over the segments of all loops. A region needs at least one loop.

[IN-13] Nesting and components. The depth of a loop is the number of other loops that contain it, found by a point-in-loop test (ray crossing, with arcs) of one of its vertices against each loop whose bounding box contains it; since loops are disjoint, one vertex decides. A loop of even depth is an outer loop; a loop of odd depth is a hole of the loop of depth one less that directly contains it. Each outer loop with its holes forms one component of Ω, and components are numbered in the order of their outer loops in the input. Depth is unbounded: an island in a hole is the outer loop of a further component, which may have holes of its own.

Normalisation

1. **[IN-07]** Translate the centre of the bounding box of all loops to the origin and scale by $1/L$, L the half-diagonal, so all coordinates lie in $[-1, 1]$. All tolerances are relative to this unit frame; output is mapped back by the inverse transform, radii by L.
2. **[IN-08]** Orient every outer loop counter-clockwise and every hole clockwise (signed area via the shoelace formula plus the arc segment areas), so Ω is on the left of every loop. The orientation of the input loops is ignored; a reversed loop is reported in the prepared region, as in v1.
3. **[IN-09]** Merge consecutive collinear lines and consecutive co-circular arcs (same centre and radius within $\varepsilon_{\text{geom}}$). This removes spurious transition vertices.
4. **[IN-10]** Snap near-tangent joins ($|\varphi| \le \varepsilon_{\text{ang}}$) to exact tangency and record them as G^1.

IN-09 and IN-10 work on each loop separately, and never across loops.

[IN-14] Prepared region. The result of preparation keeps, for every loop, its segments, joins and merge sources as in v1, plus its input index, its depth, its component, and whether it was reversed; and for every component, its outer loop and its holes.

[IN-11] Tolerances (defaults, unit frame, all configurable)

Name	Default	Meaning
$\varepsilon_{\text{geom}}$	10^{-10}	coincidence of points and of site parameters
ε_{len}	10^{-9}	minimum segment length
ε_{ang}	10^{-9} rad	tangent and cusp detection
ε_t	10^{-11}	event-time clustering

4. Output specification

The MAT is returned as an embedded graph whose edges are parametrised by radius: every edge is the trajectory of one shock over a time interval, so r increases monotonically along it from its birth vertex to its death vertex, and $r(p)$ is exact everywhere.

Graph

- **[OUT-01]** Vertices: position p, radius r, kind (section 2), component, and the list of contacts (site id plus foot point). Incident edges are stored in counter-clockwise order; an edge that starts and ends at the same vertex appears twice.

- **[OUT-02]** Edges: birth vertex v_0 at r_0, death vertex v_1 at $r_1 > r_0$, component, and the two sites (left, right) seen from the direction of travel. Plateau edges ($r_0 = r_1$) are flagged and parametrised by arc length instead. The two sites of an edge may lie on different loops.
- **[OUT-03]** Global: the site table (original loop and segment index, kind, geometry), the normalisation transform, and per-run statistics (event counts by type, loop merges, clusters resolved, maximum residual).
- **[OUT-11]** Components: for each component of Ω, its outer loop and its holes (input loop indices), and the vertices and edges of its axis. Vertices and edges are stored component by component, in component order.

[OUT-04] Edge geometry. Each edge stores its conic type, the data needed to evaluate it in closed form, and a branch selector fixed at birth:

$$p(r) = \mathrm{shock}(A, B, r), \quad r \in [r_0, r_1], \qquad f_A(r) = p(r) - r\,\nabla d_A\big(p(r)\big)$$

f_A and f_B are the contact (foot) points on the two sites; the inscribed disk of radius r at $p(r)$ touches the boundary exactly there.

[OUT-05] Evaluation API. For every edge: position, radius, unit tangent, both feet at a given r; arc-length sampling; adaptive flattening to a polyline within a chord tolerance; and exact export as rational quadratic Bézier arcs, with r supplied as a separate per-edge function since it is not polynomial in the Bézier parameter. An edge is exported as one arc per piece that turns by less than a right angle, so every weight is positive; a piece's control point is where its end tangents meet, and its weight follows from one interior point. Flattening subdivides in the edge parameter u until the middle and quarter points of every piece lie within 0.7 of the chord tolerance of their chord.

[OUT-06] Formats. C++ object graph (primary), JSON (components, sites with their loop, vertices with their component and contacts, edges with their conic kind, radii, Bézier arcs and optional flattened polylines, inward offsets as bulge loops, and the run statistics), and SVG for inspection, showing every boundary loop, MAT edges coloured by r, inward offsets and optional inscribed disks.

Guarantees (checked by the validator in section 12)

- **[OUT-07]** For every sampled point p on an edge, $|d(p) - r| \le \varepsilon_{\mathrm{res}}$ and at least two distinct feet exist, except at degree-1 vertices.
- **[OUT-08]** The axis of each component is connected and has cycle rank equal to its number of holes ($E - V + 1 = h$ for that component), so a component without holes gives a tree, as in v1. Every hole is enclosed by a cycle of the axis that encloses no other hole.
- **[OUT-09]** The union of the inscribed disks reproduces Ω to within the sampling tolerance.
- **[OUT-10]** Same input, same output, bit for bit, on the same platform and build.

5. Algorithm: wavefront propagation

The front is a set of cyclic doubly linked lists of elements and vertices, advanced from event to event by a priority queue keyed on time; between events nothing changes combinatorially and

every vertex moves on a known conic, so the simulation is exact up to root evaluation.

[ALG-02] Front data structure

- Element: the site it offsets, prev and next vertex, and a version counter bumped whenever either neighbour changes. Its live part at time t is the arc of the offset between its two vertices.
- Vertex: kind (shock or regular), left and right element, birth time and point, branch selector, and for shocks the id of the MAT edge being traced.
- Loop: entry element and element count. A loop with no vertices (a full circle offset) is allowed. Every element carries the id of its loop; loops of one component share one engine and one queue.
- Pools: elements, vertices and events live in index-addressed vectors with free lists; ids are stable 32-bit integers, which keeps the structure cache-friendly and makes runs deterministic.

[ALG-03] Initialisation

1. Build sites from every normalised loop of the component; classify each corner (section 2).
2. Create one front loop per boundary loop, with one element per site in boundary order, inserting a zero-length point-site element at every reflex corner.
3. Create a vertex at every junction: a shock at convex corners (its MAT edge starts at a corner vertex, $r = 0$), a regular vertex at tangent and reflex joins.
4. Schedule the local event of every element and seed non-local candidates (section 6).

[ALG-04] Main loop

```
while (auto ev = queue.pop_valid()) {          // skips events whose version stamps are stale
    Cluster c = gather_cluster(*ev, queue);   // all valid events within eps_t and eps_geom of ev
    if (c.size() == 1) handle_simple(*ev);    // section 6 handlers
    else               resolve_cluster(c);    // section 8, generic topological resolution
    for (Element e : touched_elements)        // only elements whose neighbourhood changed
        reschedule(e);                        // local event plus non-local candidates
}
finalize_open_edges();                        // all loops empty; every MAT edge has both ends
```

[ALG-05] Termination. Every event creates at least one MAT vertex and the MAT has $O(n)$ vertices, so there are $O(n)$ events. The simulation of a component ends when every loop is empty, at $t = \max r$, the radius of its largest inscribed disk. With holes the axis still has $O(n)$ vertices, so the bound holds.

[ALG-06] By-product. Snapshotting the front at any t yields the exact inward offset curve of Ω at distance t as lines and arcs, at no extra cost: the engine processes every event up to t and writes each loop's live elements between their vertices' positions as a bulge loop with Ω on its left: counter-clockwise where the offset bounds the material from outside, clockwise where it bounds a hole. Loops are grouped by component. Pieces shorter than ε_{len} are dropped, so the offset at $t = 0$ is the boundary and past the largest inscribed radius it is empty.

[ALG-07] Components. Each component of Ω runs in its own engine, in component order, in the unit frame of the whole region: fronts of different components are separated by boundary

and never meet. Their results are concatenated per OUT-11. A failure in one component fails the run and names the component.

6. Events and handlers

Three generic event types produce every MAT vertex: a local collapse, a non-local split, and a non-local contact; everything else (loop annihilation, high-degree junctions, plateaus) is a coincidence of these and goes to cluster resolution in section 8. Below, $A \mid B$ denotes the front vertex between elements A and B.

Event	Trigger	Time from	MAT vertex	Front update
[EV-01] E1a collapse, shock + shock	element B between $A \mid B$ and $B \mid C$ shrinks to a point	3-site solve (A, B, C)	junction, 3 contacts	remove B; new vertex $A \mid C$
[EV-02] E1b collapse, shock + regular	shock $A \mid B$ reaches the junction of B with its tangent or reflex neighbour C	bisector (A, B) meets the junction's normal ray	transition	remove B; shock continues as $A \mid C$, edge changes conic
[EV-03] E1c collapse, regular + regular	convex arc B shrinks to its centre ($t = R$)	closed form	curvature end	remove B; new shock $A \mid C$, edge born with $r = R$
[EV-04] E2 split	shock $A \mid B$ reaches the interior of a non-adjacent element C; with circle fronts, C can also reach a regular vertex $A \mid B$ inside its live part	3-site solve (A, B, C); for a regular vertex, the hit time of [KN-04]	junction, 3 contacts; extremum (min of r) for a regular vertex, which traces no edge	split C into C_1, C_2; new shocks $A \mid C_2$ and $C_1 \mid B$; split the loop, or merge two loops [EV-09]
[EV-05] E3 contact	non-adjacent elements A and C become tangent	common normal; t = half the gap	extremum (min of r)	split A and C; two new shocks moving apart; split the loop, or merge two loops [EV-09]

Rules shared by all handlers

- **[EV-06]** A new vertex between elements X and Y is a shock if X and Y meet at a corner, and regular if they meet tangentially (only possible when they are original G^1 neighbours).
- **[EV-07]** Every shock that dies closes its MAT edge at the event vertex; every new shock opens one there.
- **[EV-08]** A loop that drops to two elements both shrinking to one point, or to one element with no vertices (a circle reaching zero radius), is annihilated: its last vertex is an extremum (max of r) or a curvature end.

- **[EV-09]** Split or merge. E2 and E3 join two points of the front. When both elements lie on one loop, the loop splits in two; when they lie on two loops, the two merge into one. Nothing else is possible, so the loop ids of the elements decide, with no geometric test. Every split adds a loop, every merge and every annihilation removes one, and a cluster [RB-03] counts by the loops it removes and creates; a component starts with $1 + h$ loops and ends with none. The cycles of the axis are not tied to particular merges; OUT-08 checks them on the output.
- **[EV-14]** Loop ids after a merge. The merged loop keeps the id of the larger of the two loops, and the elements of the smaller are relabelled by a walk, so that a run with m merges costs $O(n \log n)$ in relabelling, as splits do [EV-09].

[EV-10] Local event scheduling. Each element keeps at most one pending collapse event, recomputed in $O(1)$ whenever its version changes. Stale events are not removed from the heap; they are discarded at pop time by comparing version stamps (lazy deletion).

[EV-11] Non-local event scheduling. Split and contact candidates, between elements of one loop or of two loops of the same component, come from a windowed broad phase that relies on the Lipschitz bound of section 2: a live front piece at time $t + \Delta$ lies within Δ of the same piece at time t.

1. A window $[t_k,\, t_k + \Delta_k]$ opens when the next queued event lies beyond the current one, at the end of the previous window. The box of every live element at t_k (its two ends, and the extreme points of an arc between them), inflated by Δ_k, goes into an R-tree.
2. For each overlapping pair of non-adjacent elements (elements on different loops are never adjacent), compute exact E2 and E3 candidates in $(t_k,\, t_k + \Delta_k]$ with the kernel, and enqueue the valid ones. A shock reaching element c moves along both of its elements, so a split is a candidate only when both of their boxes meet the box of c. Roots outside the window are left to the window that contains them.
3. Δ_0 is the median distance from an element's box to the nearest box of a non-adjacent element, or the median box diagonal when that median is zero. A window doubles when the previous one found fewer than 8 candidate pairs per live element or handled fewer events than a quarter of the live elements, and halves when it found more than 64 pairs per element.
4. When a handler creates or reshapes an element inside the window, query the current R-tree for it immediately, so no candidate is missed. A new element enters the R-tree with its box from now to the end of the window.

Cluster resolution [RB-03] and loop relabelling after a split [EV-09] use the same R-tree, so an event costs time for its neighbourhood rather than for its whole loop.

[EV-12] A debug mode (`BroadPhase::all_pairs`) replaces the broad phase with all pairs and an unbounded window; the two must produce identical event sequences, bit for bit, which is a standing regression test. Contact candidates are always formed with the lower element id first, so a pair has one orientation whichever element asked for it.

[EV-13] Validity at pop time. A candidate is accepted only if every involved element and vertex still has the version it was computed with and every contact foot lies inside the live

part of its element at the event time (closed intervals, $\varepsilon_{\text{geom}}$ slack). Feet that land exactly on a vertex are handed to cluster resolution.

7. Geometric kernel

All event and trajectory computations reduce to one small solver in the unknowns $z = (x, y, t)$. Line sites give linear equations and circle sites give quadratics that share the same quadratic form, so any three sites reduce to two linear equations plus one quadratic in one parameter.

Site equations. Squaring the circle equation gives

$$\|x\|^2 - t^2 - 2\,c \cdot x + 2\sigma R\,t + \|c\|^2 - R^2 = 0 \quad \text{(circle)}, \qquad n \cdot x - t - c = 0 \quad \text{(line)}$$

Every circle shares the quadratic part $Q(z) = x^2 + y^2 - t^2$ (the Minkowski form of Laguerre geometry), so the difference of two circle equations is linear in z.

[KN-01] Three-site solve (E1a, E2). Pick one circle as the pivot (if any) and subtract it from the other circle equations; with the line equations this gives a 2×3 linear system. Its solutions form a line $z = z_0 + s\,w$, with w the cross product of the two rows. Substituting into the pivot gives

$$Q(w)\,s^2 + b\,s + c = 0,$$

solved with the cancellation-free quadratic formula (linear when $Q(w) \approx 0$). With three lines the 3×3 system has a unique solution. Each root is accepted only if:

- $t \geq t_{\text{now}} - \varepsilon_t$, and $R - \sigma t \geq -\varepsilon_{\text{geom}}$ for every circle (squaring admits negative radii);
- the point is on the correct side of each site ($d_i(x) = t$ with the original sign);
- every foot lies inside the live part of its element at t.

[KN-02] The earliest accepted root wins. A rank-deficient system (parallel lines, concentric circles, collinear centres with equal radii) is reported as degenerate and routed to section 8.

[KN-03] Shock position (two sites, fixed t). The two offset curves intersect in at most two points, symmetric about the pair's axis (the centre line, or the perpendicular from a centre to a line). A branch selector, the sign of $u \times (p - o)$ with u the axis direction and o the axis origin, is fixed at the shock's birth. Both roots coincide only at a tangency, which is always an E3 or annihilation event, so the selector never changes along an edge. Line-line shocks are affine in t.

[KN-04] Regular vertices. $q(t) = q_0 + t\,m$, where m is the inward normal at the tangent join. The E1b time solves $d_A(q_0 + t\,m) = t$: linear for a line A, quadratic for a circle A.

[KN-05] Contact (E3). Two elements first touch along a common normal (a double normal):

Pair	Common normal	Contact time
line, line	exists only if antiparallel	$t^* = g/2$ along the whole overlap: plateau

Pair	Common normal	Contact time
line, circle	perpendicular from the centre to the line	$t^* = g/2$ along that perpendicular; up to two candidates
circle, circle	the centre line	$t^* = g/2$ along it; concentric circles give a plateau

Here g is the gap between the two sites measured along the common normal. A candidate is valid if both feet are live at t^* and the fronts approach each other just before t^* (their normals are opposed along the common normal).

[KN-06] Precision. All solves run in double. When the discriminant is small relative to its terms (near-tangency) or the 2×3 system is ill-conditioned, the root is refined by Newton iteration on the original unsquared equations in binary128, and its condition estimate is stored with the event for the clustering logic.

8. Robustness and degeneracies

Degenerate configurations are the normal case in CAD data (rectangles, slots, regular polygons), so they are not perturbed away; simultaneous events are merged into clusters and resolved by one generic topological procedure that subsumes every special case.

[RB-01] Event ordering. The queue orders by $(t,\ \text{kind},\ \text{smallest involved element id})$, then by the rest of the event's identity (its elements, its vertex and their versions), so that the order never depends on when an event was found. Ties are therefore deterministic, and no hash-ordered container is ever iterated on a path that affects output.

[RB-02] Cluster formation. When an event is popped at (t^*, p^*), every other valid event with $|t - t^*| \le \varepsilon_t$ whose point lies within the cluster radius $\rho = 10\,\varepsilon_{\text{geom}}$ of a point already in the cluster joins it (a breadth-first search over a grid of cells of size ρ, linear in the number of events). A plateau contributes its whole overlap segment or arc instead of a point. Repeats of one event are dropped; a cluster of one event goes to its simple handler.

[RB-03] Cluster resolution

1. Region. Take the cluster's points, or plateau segment, and grow it by ρ into a small region D. Its anchors are the mean of the event points, or the two ends of the plateau.
2. Kill. Every element whose live part at t^* lies inside D dies. Every shock with its position in D dies and closes its MAT edge at the cluster vertex. So does a shock whose path over $[t^* - \varepsilon_t,\ t^* + \varepsilon_t]$ passes through D: a shock between nearly parallel fronts moves so fast that rounding in t^* alone can leave it far outside D. An arc whose two ends meet within D counts as having no live part there, with the same angular allowance as in step 4.
3. Chains. The surviving front near D falls into k chains, which may belong to several loops (a hole front and the front around it meeting at a cluster); each enters D through a last surviving element e_{in} and leaves through a first surviving element e_{out}.
4. Relink. Sort the ports (e_{in} and e_{out} of every chain) counter-clockwise around D by the direction in which their front leaves the anchor, and fronts tangent there by their

curvature. Directions count as tangent within ε_{ang} plus, for each curved front of radius r, the cluster's spread over r: the anchor stands for every point of the cluster, so a tangency a cluster radius from a vertex is measured slightly off, and a link between tangent ports opens its shock along their common direction. The region near D lies counter-clockwise of an e_{out} and clockwise of an e_{in}, so each e_{in} links to the e_{out} just before it. Each link is a new vertex, shock or regular by the corner rule, and a shock opens a new MAT edge moving into the sector between the two.

5. Emit. One MAT vertex at each anchor with $r = t^*$, its contacts the union of the sites touching it; for a plateau, a constant-radius edge joins the two. The vertex kind follows from its degree: three or more edges make a junction; two make a maximum (two shocks ending), a minimum (two starting) or a transition (one of each, or a plateau and one shock); one or none make a curvature end.
6. Rebuild loops by walking the relinked lists: $k = 0$ annihilates every loop in the cluster. One relink may split some loops and merge others; the new loop ids follow from the walks, with the rule of EV-14 for which loop keeps its id.

[RB-04] The simple handlers in section 6 are the special cases $k = 1$ (collapse) and $k = 2$ (split, contact); they are cross-checked against this procedure by a debug option, `Options::resolve_all_events`, that sends every event through it, and the test suite requires the same axis either way.

Typical clusters

Shape	What coincides	Result
Rectangle	two corner shocks per end and the long-side contact	two junctions joined by a plateau edge
Regular n-gon	n collapses at the centre	one vertex of degree n
Disk	a loop with no vertices	one point
Slot (two semicircles, two lines)	both arcs reach their centres as the lines meet	plateau edge between two curvature ends
Annulus (concentric circles)	the hole front and the outer front meet everywhere	one closed plateau arc at $r = (R_o - R_i)/2$ [RB-07]
Square ring (square hole centred in a square)	the four side contacts, at once	one cycle of four plateau edges at $r = w/2$ (w the wall width), each end joined by a parabolic arc to a junction on an outer corner diagonal

[RB-07] Closed plateaus. A hole whose surrounding front is its exact offset (an annulus, a ring of constant width around a rounded or polygonal hole) meets it everywhere at once: the plateau events at t^* chain into a closed curve, $k = 0$, and both loops vanish. The cluster is resolved as in RB-03 with the closed chain as its region. The axis there is a cycle of plateau edges with a vertex wherever a contributing site changes on either loop; each such vertex has degree 2 (kind transition) unless a shock also ends there. A closed plateau with no site change, the concentric annulus, gets one vertex at the point of the plateau nearest the start

vertex of the hole's first input segment, and one plateau-arc edge from that vertex back to itself.

[RB-05] Near-degeneracy. Events closer than ε_t in time but farther than $\varepsilon_{\text{geom}}$ apart are processed separately, in order. They can emit very short edges; those are kept, since they are correct, and an optional post-processing step can contract edges shorter than a user threshold.

[RB-06] Invariants checked after every event (all in debug builds, the $O(1)$ local ones in release): loops are closed and consistently oriented; every element has a non-negative live length ($\geq -\varepsilon_{\text{geom}}$); every shock has a finite speed; MAT edges are opened and closed in pairs. A violation aborts the run with a diagnostic dump of the front (JSON and SVG) and the event history. The library never returns a MAT it knows to be inconsistent.

9. Complexity and performance targets

[PF-01] With n input segments the MAT has $O(n)$ vertices and edges, so the simulation processes $O(n)$ events; local work is $O(\log n)$ per event, and the expected total is $O(n \log n)$ for typical inputs, with an $O(n^2)$ worst case confined to the non-local broad phase.

Part	Cost	Notes
Validation	$O(n \log n)$ expected	R-tree broad phase, exact pair tests
Local events	$O(n \log n)$	$O(1)$ kernel solve plus a heap operation each
Non-local candidates	$O(n \log n)$ typical, $O(n^2)$ worst	windowed R-tree; worst case is many long, nearly parallel fronts
Cluster resolution	$O(m \log m)$ per cluster of m events	sum over clusters is $O(n \log n)$
Memory	$O(n)$	pools of elements, vertices and events; stale events dropped by a heap compaction whenever the heap has doubled since the last one

[PF-02] Targets (single thread, release build, a current desktop CPU)

Input size	Target time
1,000 segments	under 5 ms
10,000 segments	under 60 ms
100,000 segments	under 1 s

Measured in M6 (release build, one core of the CI-class cloud machine, `compute_mat` on a prepared region; the benchmark suite reports `prepare` separately):

Family (VER-07)	1,000 segments	10,000 segments	larger
Filleted star (random fillets)	17 ms	1.3 s	36 s at 50,000; fails at 100,000 (below)
Wavy outline (smooth star)	80 ms	0.6 s	10 s at 100,000
Spiky star	120 ms	15 s	10 min at 50,000
Gear	530 ms	over 15 min	17 s at 4,000

The broad phase meets the identical-sequence requirement [EV-12] but not these targets: the boxes of neighbours on the same smooth curve overlap in every window, and fronts that converge on one point (gears, regular polygons) make every pair a candidate. Closing the gap is milestone M8.

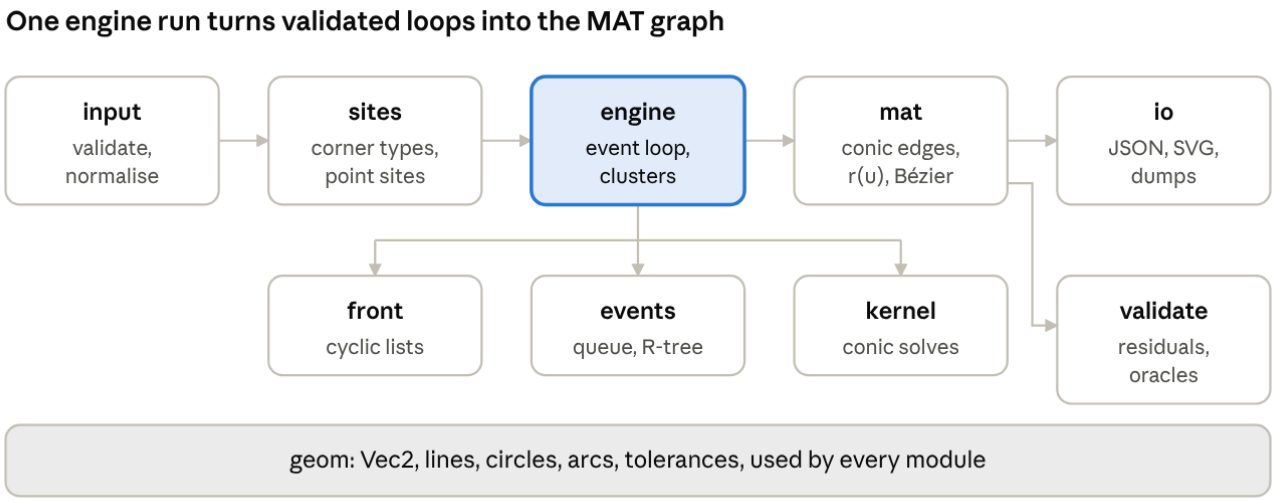
The filleted star of 100,000 segments still fails: its spikes are so thin that near-flat fillet arcs face nearly parallel sides, and the shocks between them are fast and ill-conditioned enough to leave two-element loops that never close, or links that fail RB-03. Two such cases from 40,000 and 100,000 segments are fixed and kept as regression shapes; the rest belongs to M8.

The targets are budgets for v1 to be measured against, not results; the benchmark suite in section 12 tracks them. Parallelism is out of scope for v1, because the event order is inherently sequential; batch processing of many shapes is parallel by construction, since runs share no state.

[PF-03] Scope for v2 (decided 11 October 2026). Holes are verified for correctness on inputs up to about 5,000 segments in all; slow but correct results are acceptable there. v2 sets no new speed targets and adds no stress cases beyond that size. The PF-02 targets and M8 stay where v1 left them.

10. Software architecture and C++ API

[API-01] The library, `wfmat`, is a layered set of small modules with one public entry point, `compute_mat`; the geometric kernel is pure and stateless, and all mutable state lives in one engine object per run.



wfmat modules: pipeline left to right, engine internals below

Data flows left to right; the engine owns the front and the event queue and calls the kernel for every event time and trajectory, which `mat` reuses for exact edge evaluation.

[API-02] Public API (sketch)

```
namespace wfmat {

struct Vec2 { double x, y; };
struct BulgeVertex { Vec2 p; double bulge; };          // bulge = tan(sweep / 4)
struct Loop   { std::vector<BulgeVertex> vertices; };
struct Region { std::vector<Loop> loops; };            // any order, orientation, nesting [IN-13]

struct Tolerances { double geom = 1e-10, len = 1e-9, ang = 1e-9, time = 1e-11; };
enum class BroadPhase { windowed_rtree, all_pairs };
struct Options {
    Tolerances tol;
    BroadPhase broad_phase = BroadPhase::windowed_rtree;
    bool       debug_checks = false;                   // full invariants after every event
    bool       resolve_all_events = false;             // [RB-04] every event through cluster resolution
};

enum class VertexKind { corner, curvature_end, junction, transition, extremum_min, extremum_max };
enum class ConicKind  { line, parabola, ellipse, hyperbola, plateau_line, plateau_arc };

using SiteId = std::uint32_t; using VertexId = std::uint32_t; using EdgeId = std::uint32_t;
struct Contact   { SiteId site; Vec2 foot; };
struct MatVertex { Vec2 p; double r; VertexKind kind; std::uint32_t component;
                   std::vector<Contact> contacts; std::vector<EdgeId> edges; };  // edges CCW
struct Component { std::uint32_t outer; std::vector<std::uint32_t> holes;      // input loop indices
                   VertexId first_vertex, vertex_count; EdgeId first_edge, edge_count; };  // [OUT-11]

class MatEdge {
public:
    VertexId  v0() const;  VertexId v1() const;        // birth, death; r(v0) <= r(v1)
    SiteId    left() const; SiteId right() const;
    std::uint32_t component() const;
    ConicKind kind() const;
    double r0() const; double r1() const;
    Vec2 point(double r) const;                        // not for plateau edges
    Vec2 point_at(double u) const;                     // u in [0, 1], all edges
    double radius_at(double u) const;
    Vec2 tangent_at(double u) const;
```

```
    std::pair<Vec2, Vec2> feet_at(double u) const;     // contact points on left, right
    std::vector<RationalQuadBezier> bezier() const;    // exact conic arcs, each < 90 degrees
    void flatten(double chord_tol, std::vector<Vec2>& out) const;
};

class MedialAxis {
public:
    std::span<const MatVertex> vertices() const;
    std::span<const MatEdge>   edges() const;
    std::span<const Site>      sites() const;          // maps back to input loop and segment
    std::span<const Component> components() const;
    const RunStats&            stats() const;
};

Result<MedialAxis> compute_mat(const Region&, const Options& = {});
Result<std::vector<std::vector<Loop>>> inward_offset(const Region&, double distance, const Options& = {});
                                                     // one list of loops per component [ALG-06]
// also for a PreparedRegion; std::string io::write_mat_json(mat, offsets, {units, chord_tol});

} // namespace wfmat
```

Version 2.0 changes the API in three places, so the library version moves to 2.0.0: `Region` holds a list of loops instead of `outer` (`Region{{loop}}` is the v1 region), vertices and edges carry their component, and `inward_offset` groups its loops by component.

[API-03] `Result<T>` is `tl::expected<T, Error>`, keeping the C++20 baseline (it maps directly onto `std::expected` if the project later moves to C++23); `Error` carries a code (`invalid_input`, `numerical_failure`, `invariant_violation`, or `unsupported` for valid input that the current milestone does not yet handle, with the milestone that will in the error's requirement field), the offending loop and segment (input numbering, IN-02) and component, and the path of the diagnostic dump if one was written. All output coordinates and radii are in the caller's units; normalisation is internal. A command-line tool, `wfmat-cli`, wraps `compute_mat` and `inward_offset` for JSON in, JSON and SVG out: `--json`, `--svg`, `--flatten <tolerance>`, `--offset <distance>` (repeatable) and `--version`. `docs/api.md` documents the public API.

11. Libraries and licensing

[LIB-01] The core needs only the C++20 standard library plus three header-only Boost components; everything else is confined to I/O, tests and benchmarks, and every dependency is public domain or permissively licensed (BSL-1.0, MIT, Apache-2.0).

Library	License	Used for	Scope
C++20 standard library	n/a	containers, `span`, `variant`, `<numbers>`	core
Boost.Geometry (`index::rtree`)	BSL-1.0	broad phase, input validation	core
Boost.Multiprecision (`cpp_bin_float_quad`)	BSL-1.0	binary128 root refinement, portable across compilers	core
Boost.Math (`toms748_solve`)	BSL-1.0	bracketed root fallback when Newton stalls	core
`tl::expected`	CC0 (public domain)	`Result` type on the C++20 baseline	core
nlohmann/json	MIT	JSON input and output	io

Library	License	Used for	Scope
Catch2 v3	BSL-1.0	unit and property tests	tests
Boost.Polygon (`voronoi`)	BSL-1.0	independent oracle for polygonal inputs	tests
Google Benchmark	Apache-2.0	performance suite	bench

[LIB-02] Deliberately not used. CGAL's Segment Delaunay Graph and Apollonius Graph packages are GPL-3.0 or commercial; OpenVoronoi is LGPL-2.1; VRONI and ArcVRONI are not openly licensed. Eigen (MPL-2.0) is unnecessary for 2×3 systems.

[LIB-03] Build. CMake 3.25 or newer, dependencies through a vcpkg manifest with a FetchContent fallback, tested on GCC 12+, Clang 15+ and MSVC 19.36+. The library compiles with `-ffp-contract=off` (and `/fp:precise` on MSVC) and never with fast-math, because fused multiply-add changes rounding and would break bit-for-bit determinism.

12. Verification and test plan

Every test cites the requirement identifiers it verifies. Correctness is established at three levels: the kernel against closed-form answers, whole MATs against shapes with known axes and against an independent oracle, and every run against the defining property of the MAT.

[VER-01] Kernel unit tests. Every site pair and triple type (line, convex arc, concave arc, point) for the three-site solve, shock trajectories and contact times, compared with exact symbolic results (generated once with SymPy, `tools/fixtures/gen_kernel_fixtures.py`, and stored as fixtures). Near-tangent and ill-conditioned cases check the binary128 refinement.

[VER-02] Canonical shapes with known MAT

Shape	Expected axis
Disk	one point, $r = R$
Rectangle $a \times b$	four 45° segments and a plateau of length $a - b$ at $r = b/2$
Equilateral triangle, regular n-gon	n segments meeting at the centre, degree n
Slot (stadium)	one plateau edge between the arc centres, $r = R$
L-shape	reflex corner gives two parabolic edges
Rectangle with one filleted corner	the corner branch ends at the fillet centre, a curvature end with $r = R$
Annulus, radii $R_o > R_i$, concentric	one closed plateau arc, radius $(R_o + R_i)/2$, $r = (R_o - R_i)/2$
Eccentric annulus, centres e apart	the ellipse with foci at both centres and major semi-axis $(R_o + R_i)/2$; r from $(R_o - R_i - e)/2$ (minimum) to $(R_o - R_i + e)/2$ (maximum)
Square ring (square hole centred in a square)	as in the section 8 table: one cycle, four plateaus

Shape	Expected axis
Disk with k equal circular holes on a concentric circle	k-fold symmetric axis with k independent cycles
Nested islands (square ring, disk in the hole, ring in that disk …)	one component per outer loop, each with the axis of its own row above

[VER-03] Oracle comparison. For polygonal inputs, with or without holes, Boost.Polygon's Voronoi diagram of the segments of all loops, restricted to the material and with reflex-corner spokes removed, must match our axis to within 10^{-9} of the shape size. For inputs with arcs, a dense sampling oracle (Voronoi of fine polylines) must converge to our axis as the sampling refines.

[VER-04] Property checks run on every test output (the validator in section 4)

- Residual: $|d(p) - r|$ along each edge, with d computed by brute force over all sites.
- Topology: per component, connected with cycle rank equal to its number of holes [OUT-08], with degrees matching vertex kinds.
- Reconstruction: the union of inscribed disks matches Ω within the sampling tolerance.
- Invariance: translating, rotating, scaling, reversing any loop, and listing the loops in another order give the same axis up to the transform and the component numbering.

[VER-05] Randomised testing. Random star-shaped polygons with filleted corners, and random arc chains, at 10^4 shapes per CI run with fixed seeds. With holes: the same outlines with one to eight random holes (disks, small filleted polygons, scaled copies of the outline) placed by rejection so they stay disjoint, at 10^3 shapes per CI run and at most about 5,000 segments each [PF-03]. Any failure is minimised automatically and stored as a regression fixture.

[VER-06] Degeneracy suite. Families built to hit clusters: rectangles and slots, regular polygons up to $n = 1024$, cocircular point sets, gears, and shapes with exact symmetry along both axes; with holes, the closed plateaus of RB-07 (annuli, constant-width rings around rounded and polygonal holes), symmetric hole patterns, and holes whose fronts reach the outer front at a corner.

[VER-07] Benchmarks. (v1; v2 adds none, PF-03.) Google Benchmark on scaled families (gear outlines, random fillets on a star polygon, a smooth wavy outline, a spiky star polygon; a font glyph set to follow) at 10^3 to 10^5 segments, tracked against the targets in section 9, with the all-pairs mode at 10^3 for scale.

13. Milestones

Implementation proceeds in seven increments up to the v1.0 release, one more for performance (M8), and M4, holes, which is v2, each closed by an exit test, so that the hardest parts (arcs, clusters) land on a validated base; no dates are set yet.

1. **M0 Scaffold.** CMake project, geom, input validation and normalisation, JSON reader, SVG writer, brute-force distance validator. Exit: all canonical shapes load, validate and render.

2. **M1 Kernel.** Site equations, three-site solve, trajectories, contact times, binary128 refinement. Exit: kernel unit tests pass against the SymPy fixtures.
3. **M2 Polygons.** Lines and reflex point sites, events E1a, E1b, E2, E3, all-pairs broad phase. Exit: matches the Boost.Polygon oracle on 10^4 random polygons.
4. **M3 Arcs.** Convex and concave arcs, tangent joins, E1c. Exit: canonical arc shapes and the sampling oracle.
5. **M4 Holes (v2).** Multiple loops and loop merges, delivered in three parts after the v1 release, one pull request each:
   - **M4.1 Input.** `Region` with several loops, JSON `loops` (and `outer` with `holes`), IN-12 disjointness, IN-13 nesting and components, IN-08 orientation by depth, prepared region per IN-14, SVG of all loops; `compute_mat` still answers `unsupported` for a region with holes, naming M4.2. Exit: valid and invalid multi-loop inputs load, validate and render.
   - **M4.2 Engine.** Per-component runs (ALG-07), events across loops with merges (EV-09, EV-11, EV-14), clusters on several loops and closed plateaus (RB-03, RB-07), components in the output (OUT-01 to OUT-03, OUT-11) and the cycle-rank check (OUT-08). Exit: the VER-02 hole shapes and the degeneracy suite with holes, through both the simple handlers and `resolve_all_events`, and identical event sequences to all-pairs mode.
   - **M4.3 Release 2.0.** Oracle comparison and randomised tests with holes (VER-03, VER-05), offsets and JSON grouped by component, CLI, `docs/api.md`, and a pinned vcpkg baseline (`vcpkg-configuration.json`) so Visual Studio 2026's bundled vcpkg builds the repository. Exit: v2.0.0 tag.
6. **M5 Degeneracies.** Cluster resolution and plateau edges. Exit: degeneracy suite passes; simple handlers agree with cluster resolution.
7. **M6 Performance.** Windowed R-tree broad phase, heap compaction. Exit: identical event sequences to all-pairs mode. Merged with the section 9 targets still missed, which move to M8 (decided 10 October 2026).
8. **M7 Release.** Bézier export, flattening, inward offsets, CLI, API documentation. Exit: v1.0 tag, created by CI when the project version on main reaches 1.0.0.
9. **M8 Performance II (v1.x).** A candidate pre-pass that does not degrade on converging fronts (for example a Voronoi or Delaunay diagram of a boundary sampling), and better conditioning of fast shocks between near-parallel fronts. Exit: section 9 targets met, and the 100,000-segment filleted star passes.

14. Decisions and risks

The six scope decisions of v1 were confirmed on 8 October 2026 and the v2 decisions on 11 October 2026; five technical risks have named mitigations.

Decisions

Topic	Decision
Licensing	permissive licences (BSL-1.0, MIT, Apache-2.0) are acceptable; strict public domain is not required
Holes	deferred to v2; v1 accepts a single loop
Holes in v2 (11 October 2026)	loops nested to any depth; material between each outer loop and the loops directly inside it; one axis component per outer loop; loops pairwise disjoint; orientation derived from nesting; interior MAT only
Verification scope in v2 (11 October 2026)	correctness up to about 5,000 segments; slow but correct is acceptable; no new performance targets
Exterior MAT	out of scope
Language level	C++20 baseline
Input formats	bulge JSON and the C++ API only; no external readers
Cusps	absent from the data; validation still rejects them

Risks

Risk	Effect	Mitigation
Near-degenerate inputs just outside ε	wrong local topology in a cluster	degeneracy and randomised suites; debug cross-check of simple handlers against cluster resolution; diagnostic dumps
Missed non-local event in the broad phase	missing MAT branch	Lipschitz-based window inflation; all-pairs mode as a standing regression oracle
Ill-conditioned roots near tangency	event time or position error	binary128 refinement with condition estimates feeding the clustering
$O(n^2)$ non-local candidates on adversarial shapes	slow runs	adaptive window size; measured in the benchmark suite
Closed plateaus and clusters spanning two loops (v2)	wrong relinking, a loop that never closes	RB-07 handled inside generic cluster resolution; constant-width rings in the degeneracy suite; cross-check with `resolve_all_events`

15. References

Cited from memory, not re-checked online for this draft.

- H. Blum, “A transformation for extracting new descriptors of shape”, 1967: the grassfire definition of the medial axis.
- H.-I. Choi, S.-W. Choi, H.-P. Moon, “Mathematical theory of medial axis transform”, *Pacific J. Math.*, 1997: structure and finiteness of the MAT for piecewise-analytic boundaries.
- M. Held, “VRONI: An engineering approach to the reliable and efficient computation of Voronoi diagrams of points and line segments”, *Computational Geometry*, 2001; and Held

and Huber on circular arcs, *Computer-Aided Design*, 2009: robustness lessons for line and arc sites.

- D. T. Lee, “Medial axis transformation of a planar shape”, *IEEE PAMI*, 1982: the classical $O(n \log n)$ result for polygons.
- O. Aichholzer, F. Aurenhammer et al., “A novel type of skeleton for polygons”, 1995, and Eppstein and Erickson, 1999: wavefront and event structure of the straight skeleton, the polygonal analogue of this design.
- Boost.Polygon Voronoi documentation (A. Sydorchuk): the oracle used in section 12.

Appendix A. Requirements index

ID	Section	Requirement
SC-01	1	Disjoint nested loops of lines and arcs as input
SC-02	1	Full interior MAT with exact conic edges and radius
SC-03	1	Exact handling of CAD degeneracies
SC-04	1	C++20, permissive dependencies, determinism, no global state
ALG-01	2	Corner classification by turning angle
IN-01	3	Bulge-form loop representation
IN-02	3	JSON input schema
IN-03	3	Minimum segment length and arc sweep
IN-04	3	Each loop simple
IN-05	3	Withdrawn in 2.0 (was: exactly one loop)
IN-06	3	Cusp corners rejected
IN-07	3	Normalisation to the unit frame
IN-08	3	Orientation by nesting depth
IN-09	3	Merge collinear and co-circular neighbours
IN-10	3	Snap near-tangent joins to G1
IN-11	3	Configurable tolerances with defaults
IN-12	3	Loops pairwise disjoint
IN-13	3	Nesting depth and components
IN-14	3	Prepared region with loops and components
OUT-01	4	Vertex record
OUT-02	4	Edge record and plateau parametrisation
OUT-03	4	Global data: site table, transform, statistics
OUT-04	4	Closed-form edge geometry and feet
OUT-05	4	Edge evaluation API

ID	Section	Requirement
OUT-06	4	Output formats: C++, JSON, SVG
OUT-07	4	Distance residual and two feet
OUT-08	4	Per component: connected, cycle rank equals holes
OUT-09	4	Disk union reproduces the region
OUT-10	4	Bit-for-bit determinism
OUT-11	4	Component table
ALG-02	5	Front data structure
ALG-03	5	Initialisation
ALG-04	5	Main event loop
ALG-05	5	Termination
ALG-06	5	Inward offset by-product
ALG-07	5	One engine run per component
EV-01	6	E1a collapse, shock + shock
EV-02	6	E1b collapse, shock + regular
EV-03	6	E1c collapse at curvature centre
EV-04	6	E2 split
EV-05	6	E3 contact
EV-06	6	Shock or regular vertex rule
EV-07	6	MAT edges opened and closed by shocks
EV-08	6	Loop annihilation
EV-09	6	Split or merge by loop identity
EV-10	6	Local event scheduling with lazy deletion
EV-11	6	Windowed non-local broad phase
EV-12	6	All-pairs debug mode equivalence
EV-13	6	Validity check at pop time
EV-14	6	Loop ids after a merge
KN-01	7	Three-site solve and root acceptance
KN-02	7	Earliest root; degenerate systems routed to section 8
KN-03	7	Shock position and branch selector
KN-04	7	Regular vertex trajectory and E1b time
KN-05	7	Contact times along common normals
KN-06	7	Binary128 refinement of ill-conditioned roots
RB-01	8	Deterministic event ordering

ID	Section	Requirement
RB-02	8	Cluster formation
RB-03	8	Generic cluster resolution
RB-04	8	Simple handlers cross-checked against clusters
RB-05	8	Near-degenerate events processed separately
RB-06	8	Invariant checks and diagnostic abort
RB-07	8	Closed plateaus
PF-01	9	Complexity bounds
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API-01	10	Single entry point, stateless kernel
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API-03	10	Result and Error types, caller units, CLI
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LIB-02	11	Excluded libraries
LIB-03	11	Build toolchain and floating-point flags
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VER-03	12	Oracle comparison
VER-04	12	Property checks on every output
VER-05	12	Randomised testing
VER-06	12	Degeneracy suite
VER-07	12	Benchmarks